Name _____ Period _____ Date _____

5-1 Lesson Quiz: nth Roots, Radicals, and Rational Exponents short form

Question numbers match the Savvas Lesson Quiz. Questions 1 and 5 are not on this form. Show your work.

2. Select **all** expressions that are equivalent to $x^{\frac{3}{4}}$.

Work:

- ☐ **A.** $\left(x^3\right)^{\frac{1}{4}}$
- ☐ **B.** $x^{\frac{8}{6}}$
- ☐ **C.** $\left(x^{\frac{1}{4}}\right)^{\frac{2}{4}}$
- ☐ **D.** $\left(x^{\frac{1}{4}}\right)^3$
- ☐ **E.** $\sqrt[3]{x^4}$
- ☐ **F.** $\sqrt[4]{x^3}$

3. Explain what $243^{\frac{3}{5}}$ means, then evaluate it. (Circle one word in each box.)

$243^{\frac{3}{5}}$ is the [square / cube / fourth / fifth] root of 243

[squared / cubed / to the fourth power / to the fifth power].

Its value is [3 / 9 / 27 / 81].

Work:

4. Simplify the expression $\sqrt[4]{81m^{12}n^4}$. Assume all variables are positive.

Work:

- Ⓐ $9m^3n$
- Ⓑ $9m^6n^2$
- Ⓒ $3m^3n$
- Ⓓ $3m^8$

Scoring: Question 3 and Question 4 are 10 points each. Question 2 is a bonus of up to 5 points (all three correct boxes, none wrong). Total out of 20.

5-1 Lesson Quiz, short form: answer key

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2. A, D, F. A: $\left(x^3\right)^{\frac{1}{4}} = x^{\frac{3}{4}}$. D: $\left(x^{\frac{1}{4}}\right)^3 = x^{\frac{3}{4}}$. F: $\sqrt[4]{x^3} = x^{\frac{3}{4}}$ (index 4 is the denominator, power 3 the numerator). Not B: $x^{\frac{8}{6}} = x^{\frac{4}{3}}$. Not C: $\left(x^{\frac{1}{4}}\right)^{\frac{2}{4}} = x^{\frac{1}{8}}$. Not E: $\sqrt[3]{x^4} = x^{\frac{4}{3}}$ (index and power swapped). Savvas quiz item 2, DOK 2, skill A82; taught by Example 2, rehearsed by the packet's Practice. **Bonus 5:** A, D and F with no extra box; 3 for two of the three and none wrong; 0 otherwise.

3. fifth; cubed; 27. $243^{\frac{3}{5}}$ is the fifth root of 243 cubed: $\left(\sqrt[5]{243}\right)^3 = 3^3 = 27$, because $3^5 = 243$. Savvas quiz item 3, DOK 2, skill A82; taught by Example 2 (Try It 2 is the same task). **10 points:** 3 for *fifth*, 3 for *cubed*, 4 for 27 with work (243 written as 3^5, or the fifth root found first).

4. C, $3m^3n$**.** $81 = 3^4$ and $m^{12} = \left(m^3\right)^4$, so $\sqrt[4]{81m^{12}n^4} = \sqrt[4]{\left(3m^3n\right)^4} = 3m^3n$ (or raise to the $\frac{1}{4}$ power: $81^{\frac{1}{4}}m^{\frac{12}{4}}n^{\frac{4}{4}} = 3m^3n$). Check: $\left(3m^3n\right)^4 = 81m^{12}n^4$ ✓. Savvas quiz item 4, DOK 1, skill A80; taught by Example 4 (Try It 4b is the same move). **10 points:** 6 for C, 4 for work showing the radicand as $\left(3m^3n\right)^4$ or each exponent divided by 4. Common errors: A (the square root of 81 taken instead of the fourth root); B (a square root of the whole radicand: 81 to 9, every exponent halved); D (the index 4 subtracted from each exponent instead of dividing each exponent by it).

Items left off this form (available for a retake or Form B): Savvas item 1, the real fourth roots of 256 (answer 4 and -4; DOK 1, A83; Example 1, cut; the $\pm$ for an even index is Example 5B, also cut); Savvas item 5, solve $-4x^3 = 32$ (answer $x = -2$; DOK 2, A84; Example 5, cut in the fast packet: divide by -4, then $x = \sqrt[3]{-8} = -2$).

IXL replacement: SmartScore 70 or above on 9TM, KJX and LQX replaces this quiz with 20/20.